Aamir Hamid

PhD Student in Computer Science, UMBC
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PhD student working on privacy-preserving data management and generative AI, with industry experience building LLM and analytics systems at AMD and Esri.

EDUCATION

PhD in Computer Science 2021 – Present
University of Maryland, Baltimore County (UMBC) · Baltimore, MD
- Advisor: Prof. Roberto Yus

MS in Computer Science 2017 – 2020
University of Kashmir · Srinagar, India

EXPERIENCE

Data Scientist Intern 2026
Esri · Redlands, California
- Fine-tuned Gemma-3-1B with LoRA/PEFT and extended the tokenizer with 4,000 domain-specific single-token actions to model ArcGIS Pro user-event (EUEI) sequences for next-action prediction.
- Designed and trained a compact decoder-only transformer from scratch, outperforming the fine-tuned 1B model on top-3 next-action accuracy (0.582 vs. 0.549) at a fraction of the parameters.
- Deployed models for interactive inference via a Streamlit app with top-K action prediction, and merged the LoRA adapter with the extended vocabulary into a single GGUF for CPU/llama.cpp serving.
- Mined 163.8M telemetry events across 137K sessions into transition-probability workflow graphs, uncovering high-retention paths, drop-off points, and transition networks over 4,000 application events for large-scale process mining and behavioral analysis.

Data Scientist Intern 2025
AMD · Austin, Texas
- Built an agentic RMA analytics system (Smol Agents + Snowflake Cortex) to automate trend analysis and root-cause slicing over manufacturing data.
- Productized a Snowflake semantic layer across 15+ curated tables (metrics, joins, row-level access), enabling Cortex Analyst NLQ and reducing recurring turnaround by 85%.
- Developed a Cortex-based log triage pipeline to extract and summarize errors; optimized prompts to reduce token spend while maintaining extraction quality.
- Deployed production Streamlit applications for KPI exploration and AI-assisted insights used by leadership stakeholders.
- Standardized GenAI workflows using an internal evaluation process (safety/PII/latency), improving iteration speed across 20+ team members.

Data Scientist Intern 2024
AMD · Austin, Texas
- Improved supply-chain forecasting accuracy by 15% via feature engineering, validation design, and model iteration on historical operational signals.
- Eliminated redundant features and remediated data leakage in training/validation pipelines to improve deployment reliability.
- Engineered automated batch scoring pipelines (Dataiku → Snowflake) enabling monitoring of predictions and model health.

Graduate Research Assistant 2023 – Present

DAMS Lab (Prof. Roberto Yus), UMBC · Baltimore, MD

- Designed GenAIPABench, a benchmark for evaluating generative-AI privacy assistants on policy questions, summaries and citations.

PUBLICATIONS

- **GenAIPABench: A Benchmark for Generative AI-based Privacy Assistants**. Aamir Hamid, Hemanth Reddy Samidi, Tim Finin, Primal Pappachan, Roberto Yus. *Privacy Enhancing Technologies*, 2024. doi:10.56553/popets-2024-0081 · arXiv:2309.05138

SKILLS

Languages & ML	Python, SQL, PyTorch, Hugging Face, LoRA/PEFT, llama.cpp
Data & platforms	Snowflake, Snowflake Cortex, Dataiku, Streamlit, Git, Docker
Research	LLM evaluation, benchmark design, privacy policies, experimental design, scientific writing

RESEARCH INTERESTS

Privacy & data management · Large language models & generative AI · Agentic AI systems · Applied machine learning

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